

Legal & Regulatory Compliance

Legal & compliance · draft for counsel review

Program	ATLAS Habitat Initiative
Status	Draft v0.1 — for review by qualified counsel
Scope	Canada (launch) !' United States !' international/ humanitarian
Companion docs	Business Plan · Budget & Fundraising · Pitch Deck

****Disclaimer.**** This document is an internal planning framework, ****not legal advice.**** It must be reviewed and adapted by qualified counsel in each operating jurisdiction before reliance. ATLAS is ****operationally independent, not legally exempt**** — every building code, food-safety rule, and health-data law applies in full.

1. Guiding principle

ATLAS’s “decentralization” is **operational, not legal or regulatory**. We provision our own energy, water, and partial food, but we comply with — and engage early with — every applicable authority. We frame the project to regulators as **resilience, not sovereignty**. This framing is a funding and approval constraint, not a constraint.

2. Corporate structure (decision needed)

We are evaluating three structures. The choice drives tax treatment, donor incentives, investor rights, and governance.

Option	Pros	Cons	Best when
Nonprofit / registered charity (CA: CRA registered charity; US: 501(c)(3))	Issues tax receipts; eligible for foundation grants; mission-locked	Limits equity investment; restricted commercial activity; slower	Pure humanitarian deployments
B-Corp / for-profit social enterprise	Takes equity + impact capital; agile; can license OS/Kit	No charitable receipts; donor-grant access narrower	The software + Kit business
Hybrid: Foundation + operating company (recommended)	Foundation holds mission + receives charity/grants + owns/ endows the assets; OpCo builds the OS/Kit and earns recurring revenue; clean separation	More structure to run; related-party governance must be arms-length	Scaling a blended-capital venture

Recommendation: a **hybrid** — a charitable foundation (asset endowment, grants, humanitarian deployments, impact reporting) plus a B-Corp operating company (ATLAS OS SaaS, the Kit, EPC). Requires careful related-party governance and transfer-pricing so

charitable funds are not used to confirm with counsel & tax

advisor before P1 close.

3. Building, construction & zoning

Area	Requirement	ATLAS approach
Building codes	National Building Code of Canada + provincial codes (e.g. OBC in Ontario); IBC-based codes in the US	Design to code from day one; the over-provisioned physical layer is code-compliant, not a workaround
3D-printed structure	Limited code precedent; needs alternative-solution / engineered approval and stamped structural review	Treat 3D printing as a shell accelerator; foundation/roof/MEP conventional; engage AHJ early with a Professional-Engineer-stamped alternative solution
Zoning	Residential + the unusual mixed-use of farm/energy/water floors	Pre-application meetings; frame food/energy floors as building services, not industrial use; pursue variances where needed
Fire & life safety	Egress, fire separation, sprinkler/standpipe; biogas storage is a fire-code item	Deterministic, safety-rated controls (BMS/PLC) hold life safety — never gated on AI/cloud; biogas handling engineered to code with detection + isolation
Elevators / accessibility	Provincial elevator codes; accessibility (e.g. Accessible Canada Act, AODA, ADA in US)	Designed in; not bolted on
Energy systems	Electrical code; grid-interconnection / net-metering agreements; battery storage codes	Licensed electrical design; utility interconnection even when grid is backup-only

4. Water, waste & food (the differentiators carry the heaviest regulation)

System	Key regimes	Notes
Potable water	Provincial drinking-water regulations; treatment + testing/permitting	Self-supplied water is more regulated, not less — multi-stage filtration with monitored, logged water quality
Greywater / blackwater reuse	Plumbing code; provincial wastewater & reuse approvals	~90%+ reuse must meet reuse-class standards; non-potable reuse clearly separated
Anaerobic digester / biogas	Environmental permits; fire code; emissions	Engineered containment, gas detection, pressure relief
Aquaponics / produce	Food-safety (e.g. CFIA, provincial public health, Safe Food for Canadians Regs)	Commercial food handling/sale triggers inspection + traceability
Micro-livestock (eggs) & insect protein	Animal welfare; zoning; food-safety; novel-food rules for insects	Kept isolated; never open livestock farming in a tower (odor/disease/welfare/zoning)

5. Health & clinic data (Commons & Clinic floor / telehealth)

- **Canada: PHIPA** (Ontario health info) and **PIPEDA** (private-sector personal data) govern any clinic/telehealth data. Health data is high-sensitivity.
- **Telehealth:** clinician licensing per province/state; cross-jurisdiction practice rules; consent and record-retention requirements.
- **Approach:** minimize collection; encrypt at rest and in transit; strict access control and audit logging; clear consent; data-retention schedule. Telehealth uses vetted, compliant video APIs.

6. Resident data, privacy & surveillance

- **Data dignity is a product feature, not an afterthought.** Minimize collection; residents can see and control their own data.
- **Common-area cameras & occupancy sensing** governed by clear, posted policy and tenant-privacy law; no surveillance of private dwellings.
- **Sub-metering** per unit is operational telemetry, not behavioral surveillance — scoped and disclosed.
- **Legal basis:** PIPEDA (CA) / state privacy laws (US) / GDPR-class rules where applicable internationally.

7. Cybersecurity & building-automation security

Building-automation systems are a known ransomware/botnet target — **threat-model at design time, not after an incident.**

- **Network segmentation:** the building-automation network is isolated from tenant networks. A compromised tenant device must never reach lock or life-safety controllers.
- **Local fail-safe:** the building stays safe and functional during internet/grid loss. A front door that won't open during an ISP outage is a liability, not a bug.
- **Auditability:** SR&ED/CRA-grade logging of system decisions and R&D experiments.

8. AI governance & liability

- **The AI never holds a life-safety function.** A failed model, lost network, or bad optimization, never safety. This is a code-grade architectural, and it is central to our liability posture.
- **Human-in-the-loop** for any safety-critical action; the AI proposes, humans approve within pre-authorized, bounded envelopes.
- **Emerging AI regulation:** track Canada's AIDA (proposed) / provincial rules, the EU AI Act for international deployments, and sector guidance. Maintain model documentation, decision logs, and incident reporting.

9. SR&ED & R&D tax-credit compliance (a funding edge)

The forecasting, multi-agent optimization, and anomaly-detection work is plausibly **SR&ED-eligible experimental R&D** — non-dilutive funding most housing projects lack.

- **Maintain from day one:** hypothesis !' experiment !' outcome logs; time tracking on eligible R&D; technical-uncertainty documentation; segregation of eligible vs routine engineering.
- This is CRA-grade record-keeping — a transferable compliance discipline. Get a SR&ED

specialist to scope the claim before P1.

10. Fundraising & securities compliance

Capital type	Compliance considerations
Charitable donations / grants	Charitable-receipt rules (CRA / IRS); grant-agreement covenants; restricted-fund accounting
Equity (B-Corp / OpCo)	Securities law; exemptions for private placements; investor accreditation; cap-table/governance
Crowdfunding	Securities crowdfunding rules (CA provincial / SEC Reg CF in US) — disclosure + limits
Carbon credits	Verification/registry standards; additionality; avoid double-counting
Cross-border philanthropy	Qualified-donee rules; international grant-making compliance

See [Budget & Fundraising](#) for the capital stack itself.

11. International & humanitarian deployments

- **Data residency:** each destination's data-protection rules apply; resident/clinic data may not leave certain jurisdictions. ATLAS OS supports per-region data residency.
- **Local building/health/food law** governs in each country — no assumption of Canadian rules carrying over.
- **Humanitarian context:** coordinate with host-government and agency frameworks (e.g. UNHCR shelter standards); import/customs for the Kit; local labor and licensing.

12. Compliance roadmap by phase

Phase	Legal priorities
P0	Choose corporate structure; engage construction/zoning + SR&ED counsel; draft data-privacy policy
P1 (pilot)	AHJ alternative-solution approval for 3D-print shell; water/food permits; first SR&ED claim setup; donor grant agreements
P2 (ATLAS 01)	Full code compliance + occupancy; PHIPA/PIPEDA for clinic; tenant agreements + privacy disclosures
P3 (Kit)	OS licensing terms; IP protection; repeatable approvals playbook
P4 (expansion)	US + international counsel; per-region data residency; securities compliance for scaled fundraising

13. Open legal questions

1. Final corporate structure (nonprofit vs B-Corp vs hybrid).
2. Jurisdiction-specific approval path for the 3D-printed structural shell.
3. SR&ED claim scope and specialist engagement.
4. Securities structure for blended equity + philanthropic capital.
5. First-site province (drives the specific code/health/food regime).

Reviewed claims and structures herein must be validated by qualified legal and tax professionals in each operating jurisdiction prior to reliance.

